

JFET: (Junction field effect transistor)

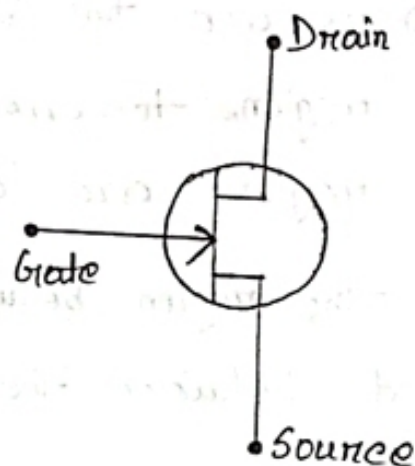
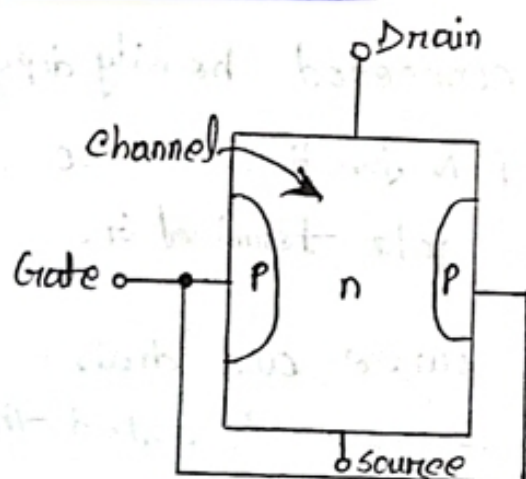
It is a three terminal semiconductor device in which current conducted by one type of carrier by electron or hole.

It is one of the simplest types of field transistor.

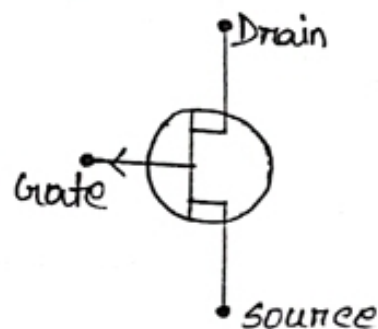
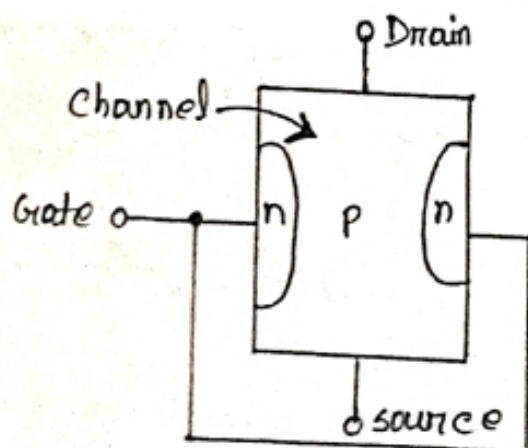
Types of JFET: n channel JFET
p channel JFET

Three terminals are: ① Drain (D)
② Gate (G)
③ Source (S)

Symbol of JFET:



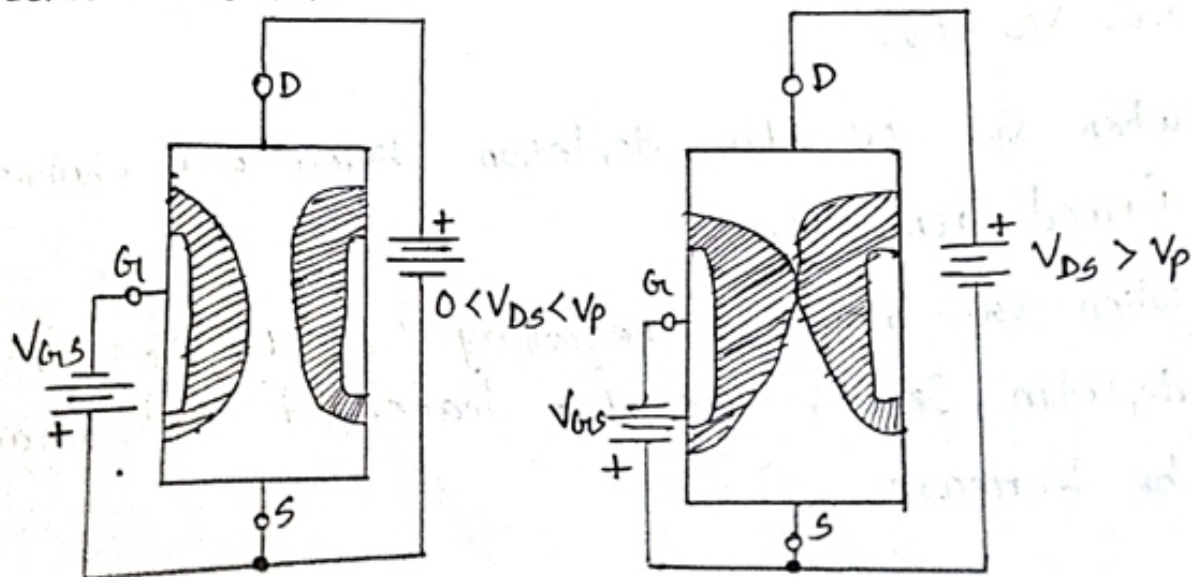
n-channel JFET



P-channel JFET

② When drain source voltage (V_{DS}) is applied at constant gate-source voltage (V_{GS}).

- ⇒ Now reverse bias at the drain end is larger than source end and so the depletion layer is wider at the drain end than source end.
- ⇒ when V_{DS} increases i.e. $0 < V_{DS} < V_p$ depletion layer at drain end is gradually increased and drain current also increased.
- ⇒ When $V_{DS} = V_p$ the channel is effectively closed at drain end and it does not allow further increase of drain current. so the drain current will become constant.



(a) channel becomes narrower as V_{DS} is increased.

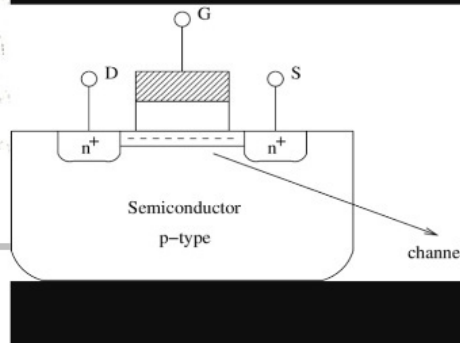
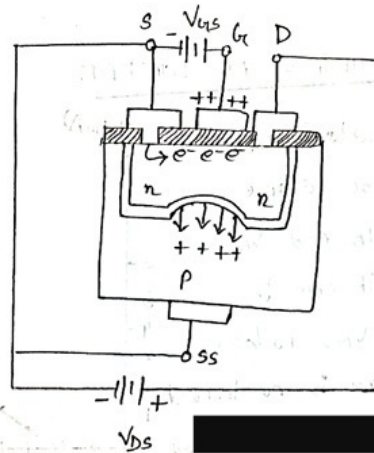
(b) Current is confined to a very narrow strip for $V_{DS} > V_p$

Feature	Enhancement-mode (E-MOSFET)	Depletion-mode (D-MOSFET)
Initial Conduction	Off (no current flow)	On (some current flow)
Gate Voltage Control	Enhances channel formation	Depletes existing channel
Gate Voltage for On	Positive (N-type) / Negative (P-type)	(Already on) Needs negative (N-type) or positive (P-type) to turn off
Typical Use Cases	Logic circuits, amplifiers	Load switches, level shifters



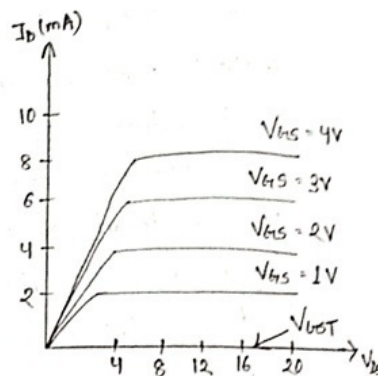
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Working Principle of a n-channel MOSFET:



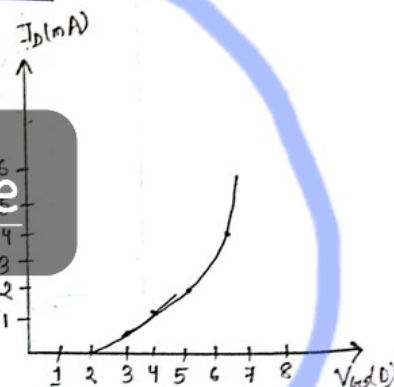
Drain characteristics of Enhancement Mosfet:

The amount of current flow is dependent on the positive amount of the potential applied at the terminal gate. If the application of the potential is below the threshold then no current flow is evident through the terminal drain. If the voltage exceeds the threshold the device gets turned ON.



Transfer characteristics of MOSFET:

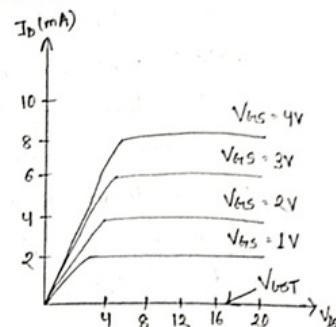
The transfer characteristics of MOSFET is the curve plotted between I_D and V_{GS} . From the curve, it is seen that, when V_{GS} is less than $V_{GS(th)}$, there is no induced channel and I_D is zero. When V_{GS} is made equal to $V_{GS(th)}$, the MOSFET is turned ON and the induced channel conducts from source to drain. The further



next page

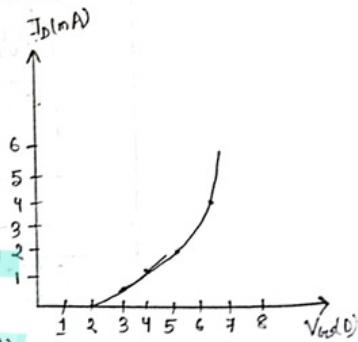
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Transfer characteristics of MOSFET:

The transfer characteristics of MOSFET is the curve plotted between I_D and V_{GS} . From the curve, it can be seen that, when V_{GS} is less than $V_{GS(th)}$, there is no induced channel and I_D is zero. When V_{GS} is made equal to $V_{GS(th)}$



The MOSFET is turned ON and the induced channel conducts from source to drain. The further

increase in the value of V_{GS} increases the width of induced channel and hence, the drain current.

Q3. A JFET has a drain current of 5 mA. If $I_{DSS} = 10$ mA and $V_{GS(off)} = -6$ V, find the value of (i) V_{GS} and (ii) V_P .

Solution.

$$I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_{GS(off)}} \right]^2$$

or

$$5 = 10 \left[1 + \frac{V_{GS}}{6} \right]^2$$

or

$$1 + \frac{V_{GS}}{6} = \sqrt{5/10} = 0.707$$

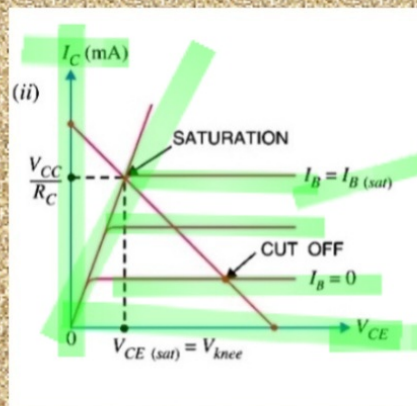
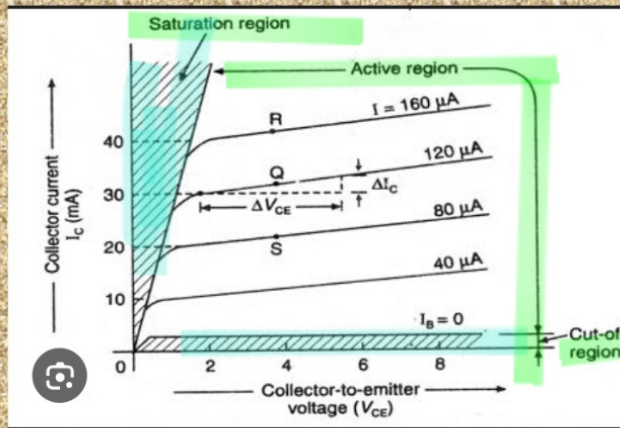
(i) $\therefore$

$$V_{GS} = -1.76 \text{ V}$$

(ii) and

$$V_P = -V_{GS(off)} = 6 \text{ V}$$

- **Cut-off (OFF):** Transistor is off, no current flows.
- **Saturation (ON):** Transistor is on, current flows freely.
- **Active:** Transistor regulates current, useful for amplification.



threshold voltage

Understand more a more balanced explanation:

A transistor switches between three regions:

- **Cut-off Region:** Acts as an open switch. With (V_{BE}) below 0.7V for silicon transistors, there's no base current, resulting in no collector current ($I_C = 0$), and (V_{CE}) is at its highest.
- **Saturation Region:** Functions as a closed switch. The base-emitter junction is forward-biased, allowing maximum base current and a large collector current, with (V_{CE}) at its lowest.
- **Active Region:** Used for amplification, not switching. The base-emitter junction is forward-biased, and the collector current is proportional to the base current, controlled by the transistor's gain.

In switching applications, a transistor is driven between the cut-off (OFF) and saturation (ON) regions, avoiding the active region.

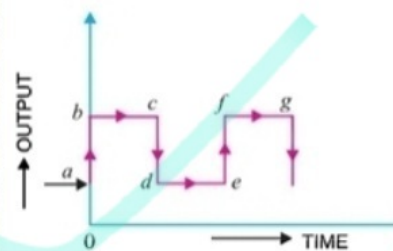
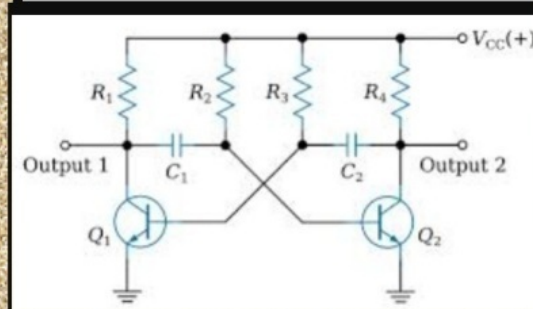
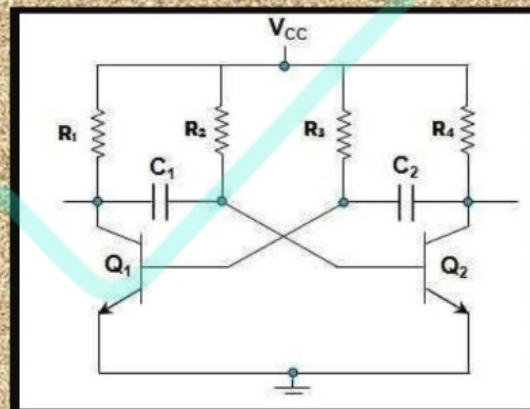


Fig. 18.14

starts charging

← ≡ Astable Multivibrator Design ...

The operation of the circuit is as follows:

1. Initially, let's assume **Q1 is OFF** and **Q2 is ON**. The capacitor **C1 charges**.
2. When the voltage across **C1 (VC1)** becomes **sufficient to turn on Q1**, **Q1 starts conducting**. This causes **Q2 to turn OFF**.
3. Now, **C2 starts charging**. When the voltage across **C2 (VC2)** becomes **sufficient to turn on Q2**, **Q2 starts conducting** and **Q1 turns OFF**.
4. This process repeats, causing the circuit to oscillate between the two states.

6. **Oscillation:** This cycle of charging and discharging the capacitor, along with the transistor switching, repeats continuously. This produces a continuous square wave output at a frequency determined by the resistors and capacitors.

Q3. Fig. 2 shows the transistor switching circuit. Given that $R_B = 2.7 \text{ k}\Omega$, $V_{BB} = 2\text{V}$, $V_{BE} = 0.7\text{V}$ and $V_{knee} = 0.7\text{V}$.

(i) Calculate the minimum value of β for saturation.

(ii) If V_{BB} is changed to 1V and transistor has minimum $\beta = 50$, will the transistor be saturated.

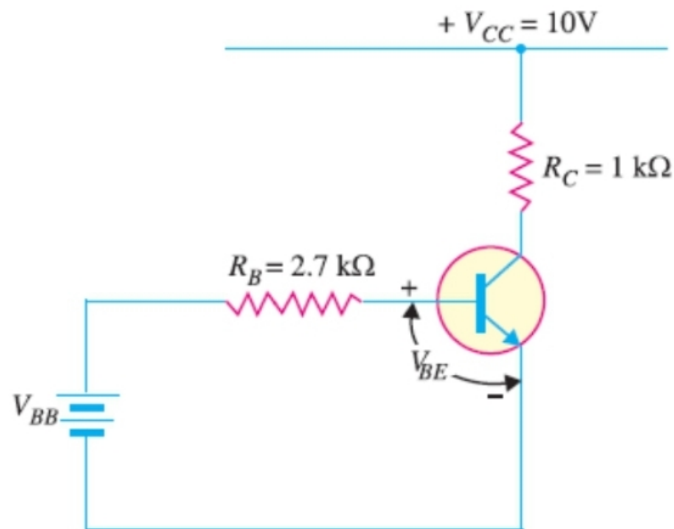


Fig.2

Solution.

$$(i) \quad I_B = \frac{V_{BB} - V_{BE}}{R_B} = \frac{2\text{V} - 0.7\text{V}}{2.7 \text{ k}\Omega} = \frac{1.3 \text{ V}}{2.7 \text{ k}\Omega} = 0.48 \text{ mA}$$

$$\text{Now } I_{C(sat)} = \frac{V_{CC} - V_{knee}}{R_C} = \frac{10\text{V} - 0.7\text{V}}{1 \text{ k}\Omega} = \frac{9.3\text{V}}{1 \text{ k}\Omega} = 9.3 \text{ mA}$$

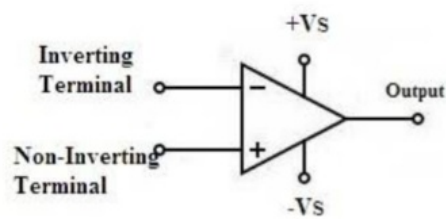
$$\therefore \text{Minimum } \beta = \frac{I_{C(sat)}}{I_B} = \frac{9.3 \text{ mA}}{0.48 \text{ mA}} = 19.4$$

$$(ii) \quad I_B = \frac{V_{BB} - V_{BE}}{R_B} = \frac{1\text{V} - 0.7\text{V}}{2.7 \text{ k}\Omega} = \frac{0.3\text{V}}{2.7 \text{ k}\Omega} = 0.111 \text{ mA}$$

$$\therefore I_C = \beta I_B = 50 \times 0.111 = 5.55 \text{ mA}$$

Since the collector current is less than saturation current ($= 9.3 \text{ mA}$), the **transistor will not be saturated.**

An operational amplifier, or op-amp, is a type of electronic device that can amplify input signals and perform mathematical operations. It has two inputs,



Operational Amplifier

Example 25.27. Find the output voltage for the circuit shown in Fig. 25.51.

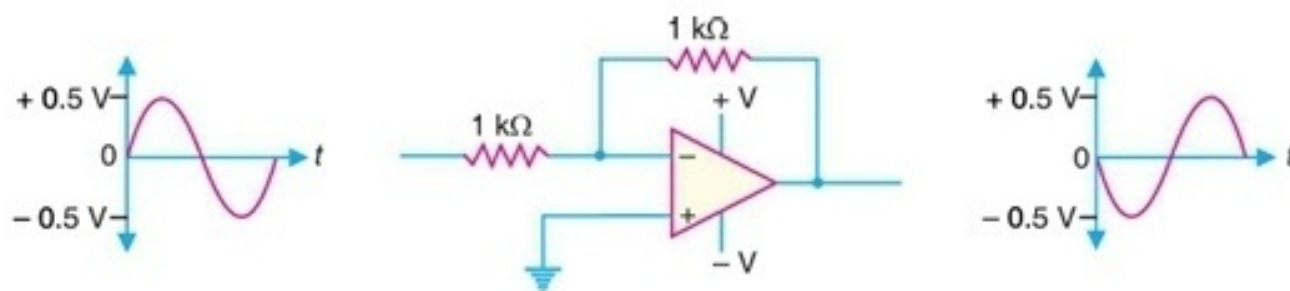


Fig. 25.51

Solution. Voltage gain, $A_{CL} = -\frac{R_f}{R_i} = -\frac{1 \text{ k}\Omega}{1 \text{ k}\Omega} = -1$

Since the voltage gain of the circuit is -1 , the output will have the same amplitude but with 180° phase inversion.

Example 25.28. Find the output voltage for the circuit shown in Fig. 25.52.

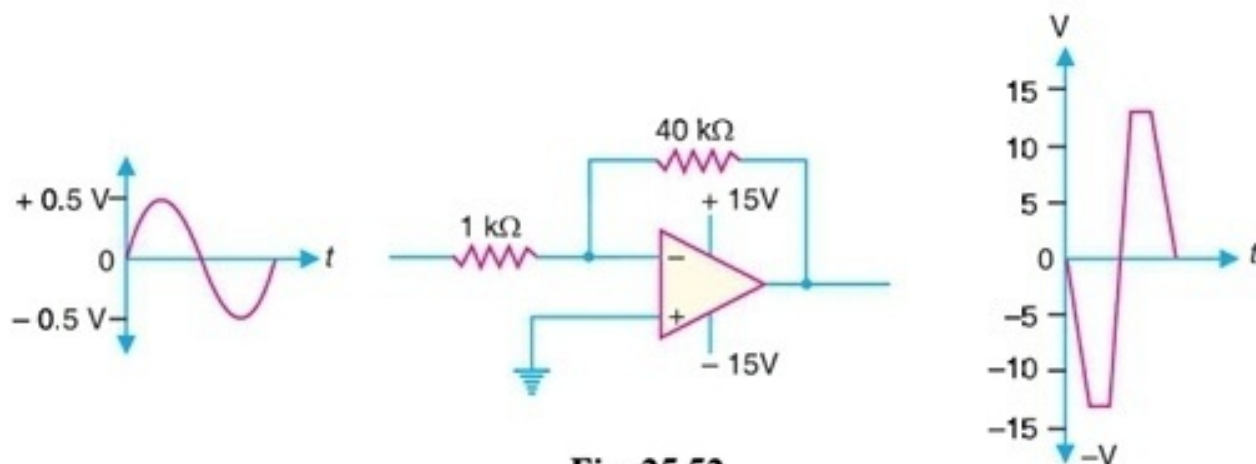


Fig. 25.52

Solution. Voltage gain, $A_{CL} = -\frac{R_f}{R_i} = -\frac{40 \text{ k}\Omega}{1 \text{ k}\Omega} = -40$

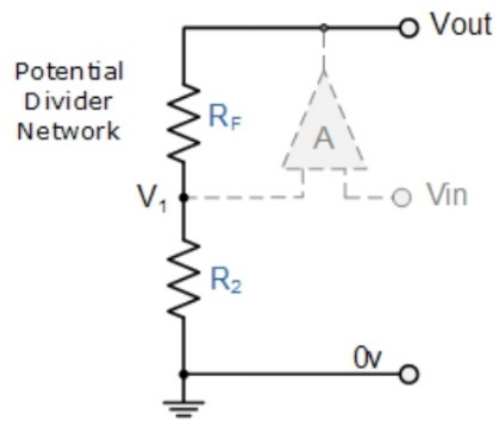
shortly and easily rewrite about
The circuit (like in example 25.27) has a much higher gain (-40) this time. This pushes the output voltage beyond its limits (saturation at $\pm 13 \text{ V}$), causing the amplifier to clip and distort the signal. However, the signal is still flipped upside down (180° phase shift).

In this instance, the circuit has a significantly increased gain of -40 . As a result, the output voltage exceeds the maximum allowable range, reaching saturation at $\pm 13 \text{ V}$. This leads to clipping and distortion of the signal. Despite this, the signal undergoes a 180° phase shift, meaning it is inverted.

Operational Amplifiers ■ 697

Note that the input signal is the same as in example 25.27 but now the voltage gain is -40 instead of -1 . Since the supply voltages are $\pm 15 \text{ V}$, the saturation occurs at $\pm 13 \text{ V}$. Since the output voltage far exceeds the saturation level, the OP-amp will be driven to deep saturation and it will behave as a non-linear amplifier. This means that the output will not have the same shape as input but will clip at the saturation voltage. Note that 180° phase inversion does occur.

Equivalent Potential Divider Network



Then using the formula to calculate the output voltage of a potential divider network, we can calculate the closed-loop voltage gain (A_V) of the **Non-inverting Amplifier** as follows:

Non-inverting Operational Amplifier Gain

$$V_1 = \frac{R_2}{R_2 + R_F} \times V_{OUT}$$

Ideal Summing Point: $V_1 = V_{IN}$

Voltage Gain, $A_{(V)}$ is equal to: $\frac{V_{OUT}}{V_{IN}}$

$$\text{Then, } A_{(V)} = \frac{V_{OUT}}{V_{IN}} = \frac{R_2 + R_F}{R_2}$$

Transpose to give: $A_{(V)} = \frac{V_{OUT}}{V_{IN}} = 1 + \frac{R_F}{R_2}$

←

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Then the closed loop voltage gain of a **Non-inverting Operational Amplifier** will be given as:

$$A_{(V)} = 1 + \frac{R_F}{R_2}$$

13.8 Effects of Negative Current Feedback

13.9 Emitter Follower

13.10 D.C. Analysis of Emitter Follower

13.11 Voltage Gain of Emitter Follower

Signal Sampling: A fraction of the output signal is sampled and fed back.

Inversion: The sampled signal is inverted (phase-shifted by 180 degrees).

Combination: The inverted signal is combined with the input signal, effectively reducing the amplitude of the input.

Stabilization: The amplifier's gain becomes less sensitive to component variations and more predictable.

input. Consequently, even a casual disturbance at the input will appear in the amplified form in the output. There is a strong tendency in amplifiers to introduce *hum* due to sudden temperature changes or stray electric and magnetic fields. Therefore, every high gain amplifier tends to give noise along with signal in its output. The noise in the output of an amplifier is undesirable and must be kept to as small a level as possible.

The noise level in amplifiers can be reduced considerably by the use of *negative feedback* i.e. by injecting a fraction of output in phase opposition to the input signal. The object of this chapter is to consider the effects and methods of providing negative feedback in transistor amplifiers.

13.1 Feedback

The process of injecting a fraction of output energy of

336 ■ Principles of Electronics

some device back to the input is known as **feedback**.

The principle of feedback is probably as old as the invention of first machine but it is only some 50 years ago that feedback has come into use in connection with electronic circuits. It has been found very useful in reducing noise in amplifiers and making amplifier operation stable. Depending upon whether the feedback energy aids or opposes the input signal, there are two basic types of feedback in amplifiers viz *positive feedback* and *negative feedback*.

(i) **Positive feedback.** When the feedback energy (voltage or current) is in phase with the input signal and thus aids it, it is called *positive feedback*. This is illustrated in Fig. 13.1. Both amplifier and feedback network introduce a phase shift around the loop, causing the feedback voltage V_f to be in phase with V_{in} .

1. **Signal Sampling:** A fraction of the output signal is sampled and fed back.

2. **Inversion:** The sampled signal is inverted (phase-shifted by 180 degrees).

3. **Combination:** The inverted signal is combined with the input signal, effectively reducing the amplitude of the input.

Amplifier gain is more stable and reliable.

becomes

The positive feedback increases the gain of the amplifier but it also increases the distortion. One important use of positive feedback is in oscillators. As we shall see in the next chapter, if positive feedback is sufficiently large, it leads to oscillations. As a matter of fact, an oscillator is a device that converts d.c. power into a.c. power of any desired frequency.

(ii) **Negative feedback.** When the feedback energy (voltage or current) is out of phase with the input signal and thus opposes it, it is called *negative feedback*. This is illustrated in Fig. 13.2. As you can see, the amplifier introduces a phase shift of 180° into the circuit while the feedback network is so designed that it introduces no phase shift (i.e., 0° phase shift). The result is that the feedback voltage V_f is 180° out of phase with the input signal V_{in} .

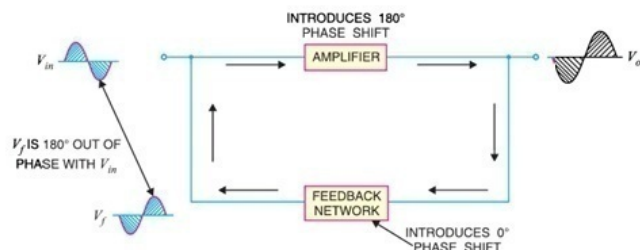


Fig. 13.2

Negative feedback reduces the gain of the amplifier. However, the advantages of negative feedback are: reduction in distortion, stability in gain, increased bandwidth and improved input and output impedances. It is due to these advantages that negative feedback is frequently employed in amplifiers.



Q3. When negative voltage feedback is applied to an amplifier of gain 100, the overall gain falls to 50.

(i) Calculate the fraction of the output voltage feedback.

(ii) If this fraction is maintained, calculate the value of the amplifier gain required if the overall stage gain is to be 75.

Solution :

(i) Gain without feedback, $A_v = 100$

Gain with feedback, $A_{vf} = 50$

Let m_v be the fraction of the output voltage feedback.

Now
$$A_{vf} = \frac{A_v}{1 + A_v m_v}$$

or
$$50 = \frac{100}{1 + 100 m_v}$$

or
$$50 + 5000 m_v = 100$$

or
$$m_v = \frac{100 - 50}{5000} = 0.01$$

(ii) $A_{vf} = 75$; $m_v = 0.01$; $A_v = ?$

$$A_{vf} = \frac{A_v}{1 + A_v m_v}$$

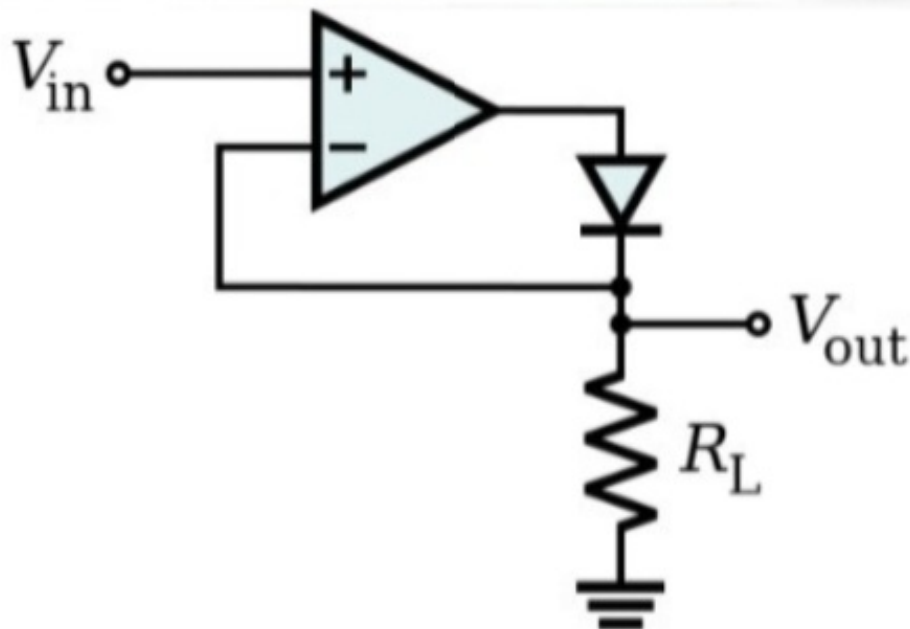
or
$$75 = \frac{A_v}{1 + 0.01 A_v}$$

or
$$75 + 0.75 A_v = A_v$$

$\therefore$
$$A_v = \frac{75}{1 - 0.75} = 300$$



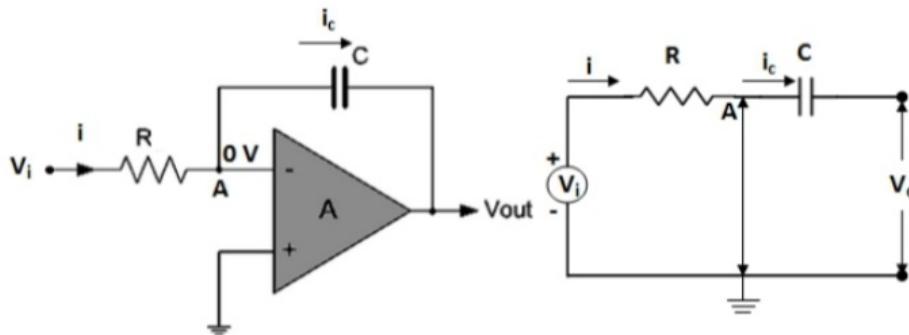
The **precision rectifier** is a configuration obtained with an **operational amplifier** in order to have a **circuit** behave like an ideal **diode** and **rectifier**.



A simple precision rectifier circuit

. When the **input voltage** is **negative**, there is a **negative voltage** on the **diode**, so it works like an **open circuit**, no current flows through the load, and the output voltage is zero. When the input is positive, it is **amplified by the operational amplifier**, which **switches the diode on**. Current flows through the load and, **because of the feedback**, the **output voltage** is **equal to the input voltage**.

Fig.4 shows the circuit of an OP-Amp integrator.



Since the input is connected to the resistor, the same current, i must pass through both the resistor and the capacitor ($i_R = i_C$) producing a V_R voltage drop across the resistor so the current, (i) flowing through this series RC network is given as:

$$i(t) = \frac{V_{IN}}{R} = \frac{V_R}{R} = C \frac{dV}{dt}$$

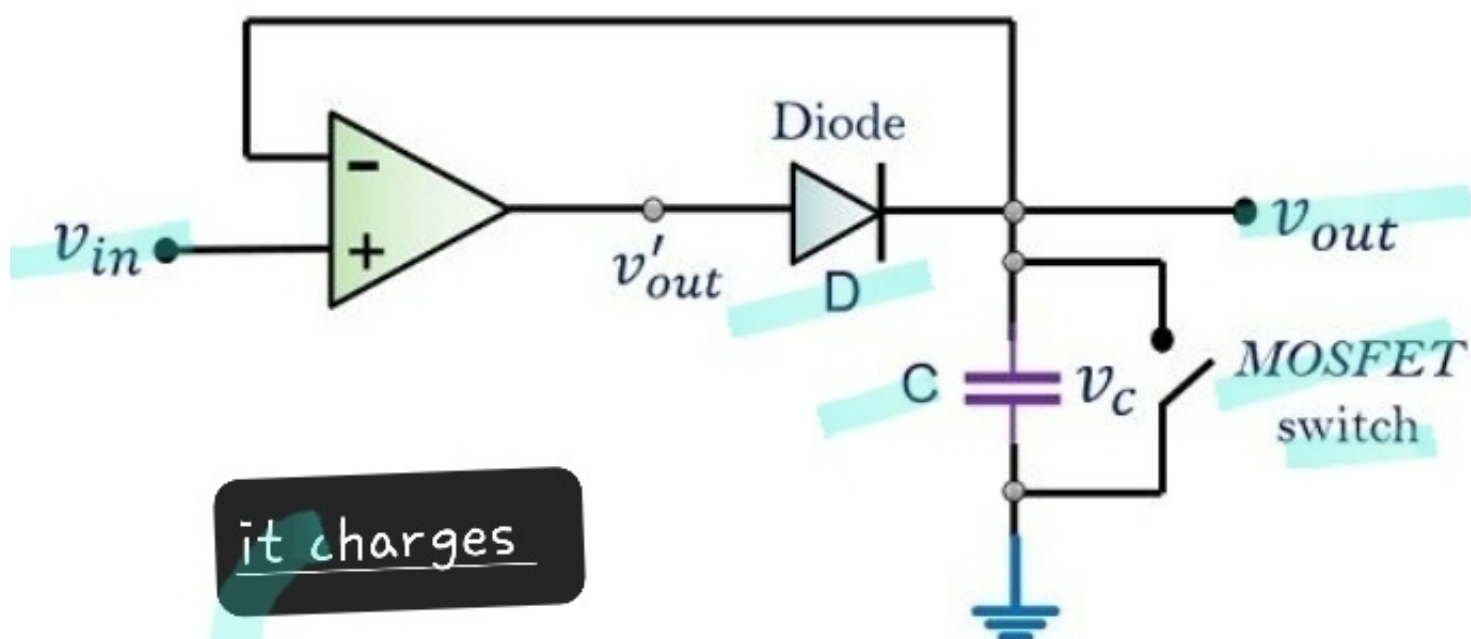
therefore:

$$V_{OUT} = V_C = \frac{Q}{C} = \frac{\int i dt}{C} = \frac{1}{C} \int i(t) dt$$

As $i = V_{IN}/R$, substituting and rearranging to solve for V_{OUT} as a function of time gives:

$$V_{OUT} = \frac{1}{C} \int \left(\frac{V_{IN}}{R} \right) dt = \frac{1}{RC} \int V_{IN} dt$$

This equation shows that the output is the integral of the input with an inversion and scale multiplier of $1/RC$.



Peak detector circuit

A peak detector circuit's main function is to identify and hold onto the highest voltage level of an input waveform. It does this through two key components: a diode and a capacitor. The diode allows current to flow to the capacitor only when the input voltage is at its maximum, **charging** the capacitor to this peak level. Once the input voltage drops, the diode stops conducting, and the capacitor holds the peak voltage, providing a steady output until a new peak is detected.

Example 25.50. Fig. 25.84 (i) shows the OP-amp integrator and the square wave input. Find the output voltage.

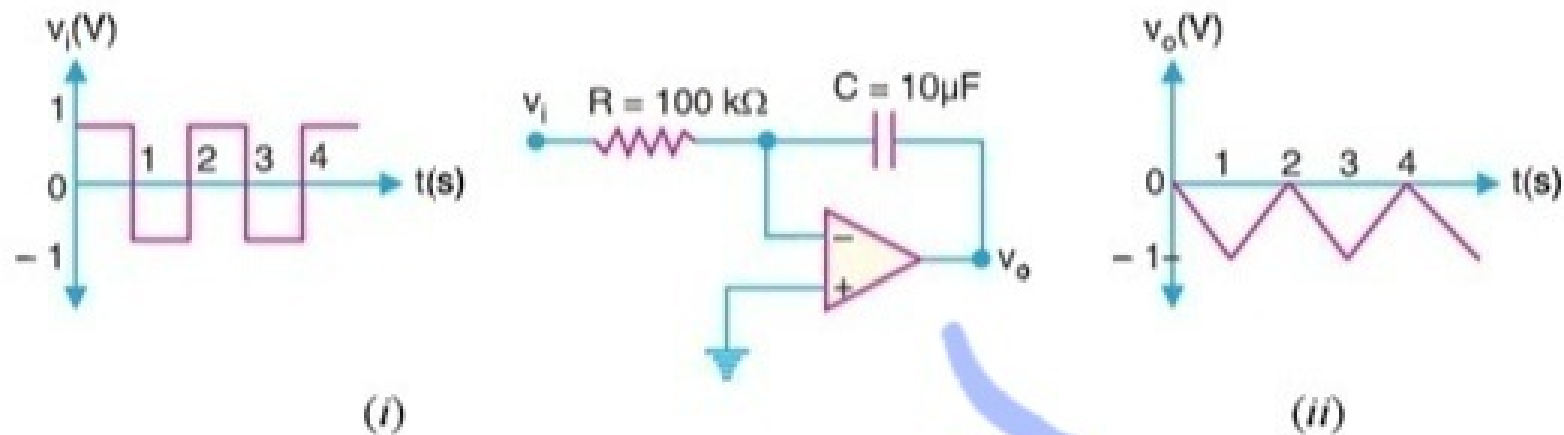


Fig. 25.84

Solution. The output voltage of this circuit is given by;

$$v_o = -\frac{1}{RC} \int_0^t v_i dt$$

Now

$$RC = (100 \text{ k}\Omega) (10 \mu\text{F}) = (100 \times 10^3 \Omega) (10 \times 10^{-6} \text{ F}) = 1 \text{ s}$$

$\therefore$

$$v_o = - \int_0^t v_i dt$$

When we integrate a constant, we get a straight line. In other words, when input voltage to an integrator is constant, the output is a linear ramp. Therefore, the integration of the square wave results in the triangular wave as shown in Fig. 25.84 (ii). Since the input to the integrator is applied to the inverting input, the output of the circuit will be 180° out of phase with the input. Thus, when the input goes positive, the output will be a negative ramp. When the input is negative, the output will be a positive ramp. Fig. 25.84 (ii) shows this relationship.

The integrator circuit inverts the input signal, resulting in an output that is the opposite phase to the input.